

2022

Cybersecurity
INSIDERS

MALWARE REPORT



SANGFOR

INTRODUCTION

Malware, and especially ransomware, are among the most destructive security threats affecting organizations of all sizes, from SMBs to large enterprises and government agencies. Malware is continuously evolving and organizations are facing significant challenges in protecting their IT environments against new types of viruses, worms, spyware, ransomware, and crypto-jacking malware.

The 2022 Malware Report was produced by Cybersecurity Insiders and Sangfor to reveal the latest malware security trends, challenges, and investment priorities.

Key findings include:

- The attack landscape is getting worse, not better. Most businesses (86%) feel an attack on their organization is likely within the next 12-months.
- A majority of organizations (72%) see remote work as a moderate to extreme risk for ransomware attacks. 39% confirm they suffered from ransomware attacks in the past, and a significant majority of 64% are not confident in their ability to detect and block an attack.
- Of those experiencing an attack, 79% think they can recover within a few days. Only 49% think they can contain the infection.
- Attacks significantly impact business and operations. 59% suffered productivity loss, followed by system downtime (41%) and data and revenue loss (28%).
- 68% of security professionals indicate they want their endpoint security to be able to block encryption. However, 86% of those attacked had to recover their data in some way after encryption.

Thanks for supporting this unique research.

We hope you find this report informative and helpful as you continue your efforts in securing your organizations against evolving malware threats.

Thank you,

Holger Schulze



Holger Schulze

CEO and Founder
Cybersecurity Insiders

Cybersecurity
INSIDERS

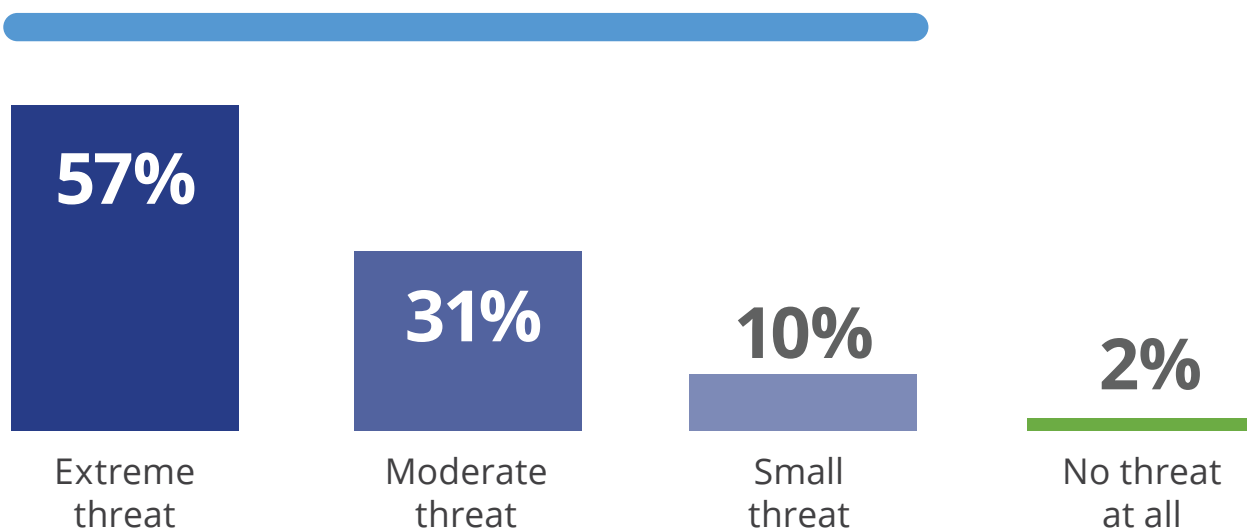
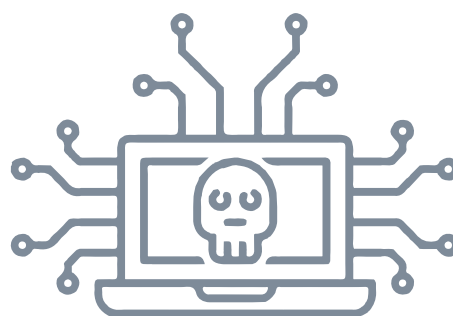
RISING THREAT LEVEL

Malware has the potential to significantly harm businesses. 98% of cybersecurity professionals agree they see malware and ransomware as a threat. The significance ranges from extreme threat (57%), moderate threat (31%), or small threat (10%).

► How significant a threat is a malware and ransomware to your business?

98%

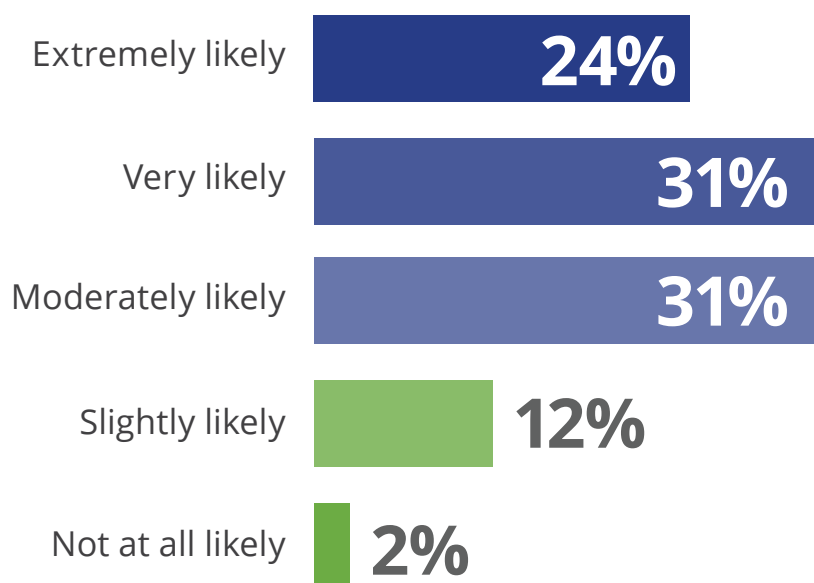
of respondents see ransomware as a threat to their business



WORSENING ATTACK OUTLOOK

The attack landscape is getting worse, not better. Most businesses (86%) feel an attack on their organization is likely within the next 12-months.

- What is the likelihood that your organization will be a target of a malware/ ransomware attack in the next 12 months?



86%

of respondents said
an attack is at least
moderately likely

REMOTE WORK RISK

With remote work becoming the new normal, malware risk remains significantly elevated. A majority of cybersecurity professionals (72%) see remote work as a moderate to extreme risk to their organization.

► How significant of a business threat are remote workers to your business?



25%



Extreme threat

47%



Moderate threat

20%



Small threat

8%



No threat at all

EXPERIENCED ATTACKS

With ransomware attacks so prevalent, it's interesting that less than half of security professionals (39%) say their organization has experienced ransomware attacks. 61% indicate they have not been affected by ransomware or aren't aware of a previous or ongoing attack.

► Has your organization suffered from ransomware attacks in the past?



RECOVERY TACTICS

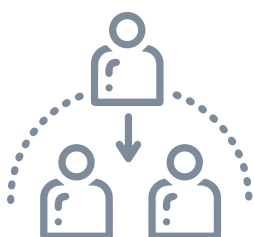
When asked how organizations recover from a ransomware attack, a majority of organizations (52%) report they re-installed or restored backups. 16% brought in third-party experts for help and 13% managed to decrypt locked files. A small percentage (5%) paid the ransom.

- If your organization suffered from a ransomware attack, how did your organization recover from the attack?



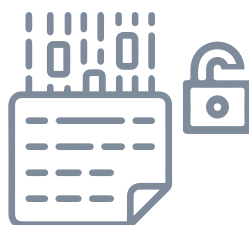
52%

Re-installed/restored



16%

Brought in
third-party



13%

Decrypted
files ourselves



5%

Paid the
ransom

Not sure/other 14%.

IMPACT OF MALWARE ATTACKS

Malware attacks impact business, operations, and IT stakeholders. Organizations highlight that attacks drive up security-related spending (59%) and productivity loss (56%). At the IT operations level, ransomware is forcing cybersecurity professionals to update IT security strategies to focus on mitigation (44%), as well as causing system downtime (41%).

- What has been the impact of malware attacks on your organization in the past 12 months?

BUSINESS IMPACT



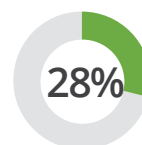
59%

Increased spending
on IT security

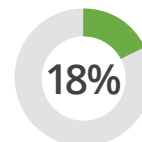


56%

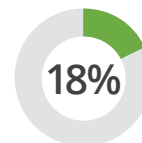
Productivity
loss



Revenue loss



Damage to
company reputation



Negative press/
bad publicity

IT OPERATIONS/SECURITY IMPACT



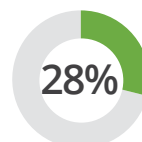
44%

Change of IT
security strategy

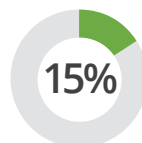


41%

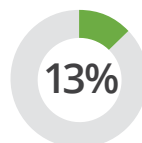
System
downtime



Data loss



Senior IT staff
(CIO, CISO) lost their jobs



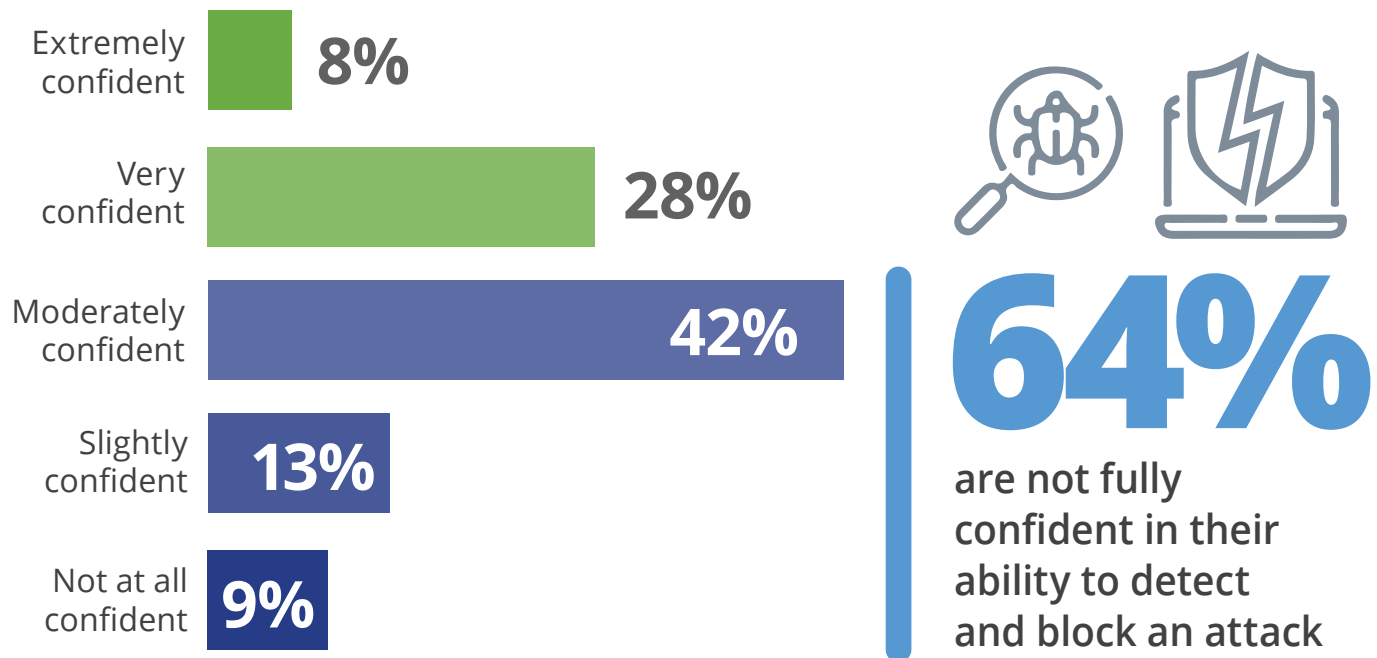
Loss of confidence
in existing cybersecurity
solutions

We did not experience any ransomware attacks 23%.

CONFIDENCE IN DETECTION

Most cybersecurity professionals (64%) do not have confidence in their organization's capacity to detect and block a malware/ransomware attack before it spreads to critical IT systems and files. Only 8% are extremely confident and 28% are very confident.

- How confident are you that your organization's defenses are capable of detecting and blocking malware/ransomware before it spreads and infects critical systems and files?



CONTAINING ACTIVE INFECTIONS

Containing active infections is essential to maintaining a solid security posture. To accomplish this goal, 49% of organizations have incident response teams and tools in place to detect and isolate the malware. 39% deploy advanced malware detection and response solutions.

- ▶ How does your organization detect and respond to lateral movement or infected computers that participate in a botnet?



49%

Incident response team
to detect and isolate



39%

Advanced, behavior-based malware
solution that protects endpoints and
has the ability to detect with automated
mitigation/remediation capability

This scenario is unlikely to happen in my organization 15% | Don't know/other 21%.

SPEED OF RECOVERY

While an attack can wreak havoc on a company's ability to operate, the majority of respondents (79%) say they need at least a few days to recover business after a ransomware attack. 10% estimate it will take more than a week to recover and another 11% of organizations believe they will never fully recover.

► How fast do you believe you can recover from a ransomware attack?

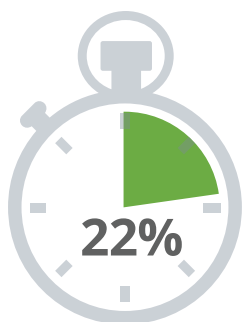


79%

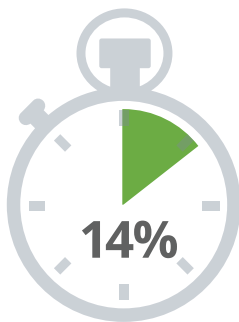
of customers need at least a few days to recover after a ransomware attack



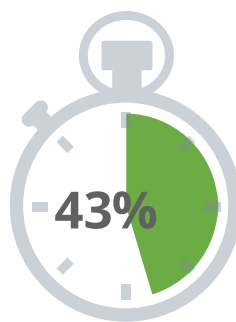
Potentially never recover



A few hours



A day



A few days



A week

RANSOMWARE DEFENSE

When asked about the primary drivers for improving ransomware defenses, most organizations want their employees to be productive, as indicated by 74% highlighting system downtime as the biggest motivator for improving their organization's ransomware defense. This is followed by mitigating the financial cost generated by ransomware attacks (71%) and the protection of sensitive business data against attacks (69%).

► What is your organization's primary driver for improving ransomware defense?



74%

Saving the organization from potential downtime



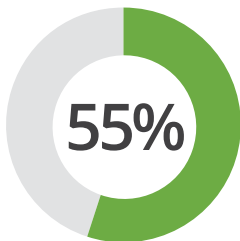
71%

Mitigating the financial costs arising from ransomware attacks

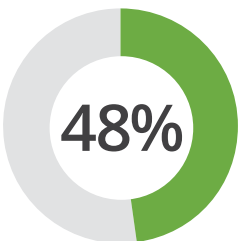


69%

Protecting confidential data related to the business and clients



Protecting the reputation of the brand



Staying a step ahead of emerging threats

Other 1%

EFFECTIVE PREVENTION

Security professionals rank anti-malware/antivirus/endpoint security (80%) as the most effective solutions in combating malware and ransomware. This is followed by user awareness and training (69%) and keeping operating systems and software current (67%) as effective ways to prevent an attack.

► What security solution(s) would you say is (are) most effective to prevent and block malware/ransomware?



80%

Anti-malware/
antivirus/endpoint
security solution



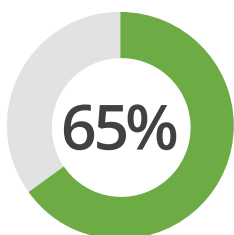
69%

User awareness
and training

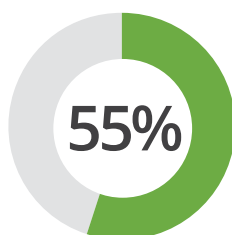


67%

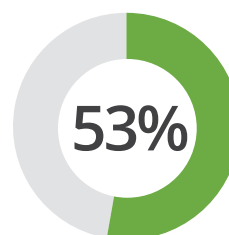
Updating/patching
operating systems
and software with
latest versions



Endpoint Detection
and Response (EDR)



Email and web
gateway



Spam filters

Infrastructure security monitoring 51% | Network IDS/traffic monitoring 50% | Internal access controls and authentication 45% | Behavior-based/machine learning endpoint protection 41% | File monitoring 36% | Sandbox 35% | User monitoring 33% | Application whitelisting 28% | Other 4%.

ESSENTIAL SOLUTION FEATURES

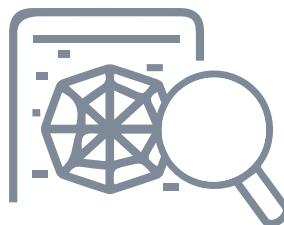
Malware protection only goes so far, organizations need to harden and patch systems in a timely fashion. Automated updating and scanning are the most critical capabilities that security professionals (76%) want in server-level malware protection solutions.

► What features do you consider most important in server-level malware protection solutions?



76%

Automatic updating
and scanning



67%

Heuristic
analysis



66%

Native file
system scanning



61%

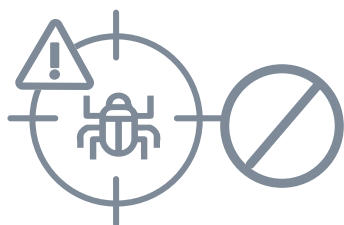
Object integrity
scanning

Other 6%.

ENDPOINT SECURITY

The increase in remote work is placing a greater emphasis on endpoint protection. Detecting and blocking malicious behavior (68%) and blocking ransomware and other attacks pre-execution (59%) rank as the most valuable endpoint security capabilities.

► What do you think is the most valuable endpoint security technology to have?



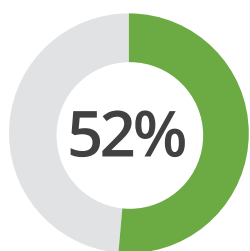
68%

Detect and block at the first sign of malicious behavior (i.e., encryption)

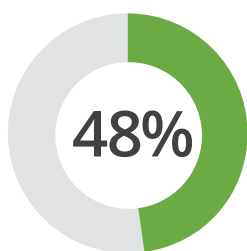


59%

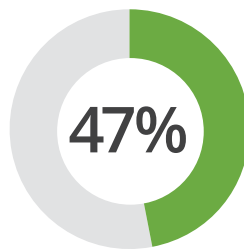
Block ransomware and others at pre-execution to stem the spread



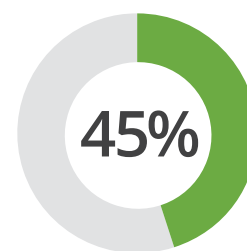
Advanced file analysis (i.e., nextgen antivirus tools)



Built-in web security preventing access to phishing, fraudulent, or exploit-hosting sites



Non-signature based detection and prevention technologies (i.e., machine learning and behavior-based)



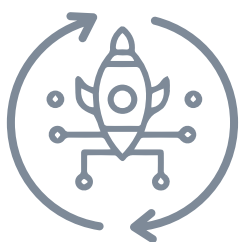
Fileless/exploit prevention through real-time behavior analysis

Endpoint integrated sandbox 44% | Automatic mitigation including the ability to roll back changes 44% | File-based detection - signature-based traditional antivirus 35% | Built-in anti-exploit 27% | Other 6%.

OBSTACLES TO DEFENSE

When asked about the biggest obstacles to malware defense, most organizations identify evolving sophistication of malware attacks (53%) and lack of budget (50%) as the primary barriers. The growing proliferation of attacks (38%) and poor user awareness (37%) is a concern as well.

► What do you believe to be your organization's biggest obstacles to improving malware/ransomware defense?



53%

Evolving sophistication
of attacks



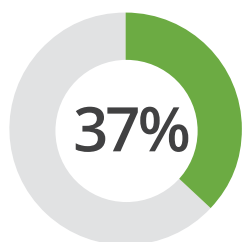
50%

Lack of
budget

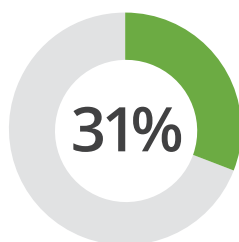


38%

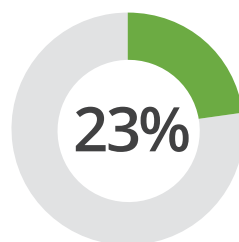
Growing proliferation
of attacks



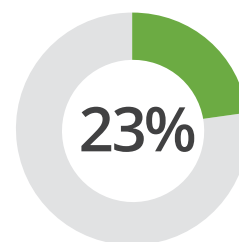
Poor user
awareness



Lack of human
resources



Lack of executive
sponsorship



Uncertainty what
security solution
to use

Our partners' lack of preparedness or response 10% | Other 8%.

SECURITY PROVIDER CRITERIA

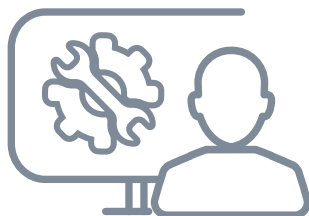
Organizations use a mix of criteria to select security providers to thwart malware and ransomware attacks. However, it is not surprising that strength of protection (74%) ranks at the top of the list. Most security professions also prioritize support (59%), cost (59%), and ease of deployment (56%) when selecting a security provider.

- What are the main criteria that you consider when selecting a security provider to protect you from malware/ransomware attacks?



74%

Strength of protection



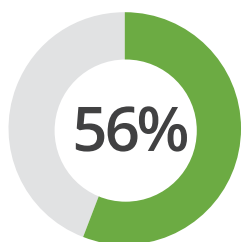
59%

Support

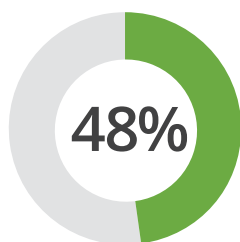


59%

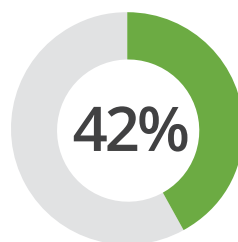
Cost



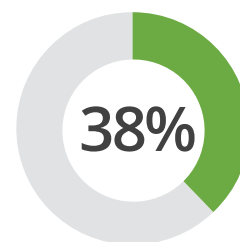
Ease of deployment



Availability of next-gen security features



Compatibility with other apps (integration)



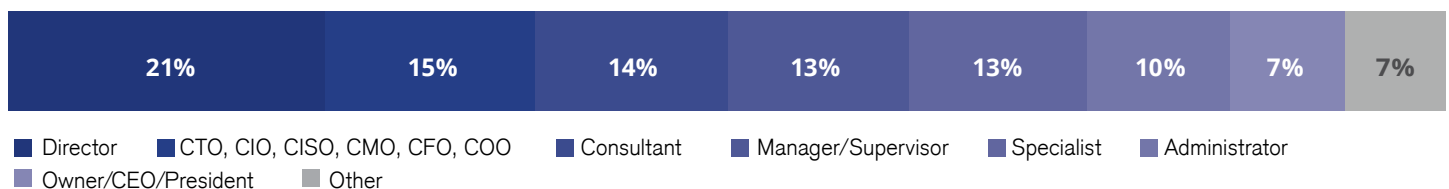
Brand reputation

False positive rate 26% | Other 2%.

METHODOLOGY & DEMOGRAPHICS

This report is based on the results of a comprehensive online survey of 231 cybersecurity professionals conducted in September 2021, to gain more insight into the latest trends, key challenges, and solutions for malware and ransomware security. The respondents range from technical executives to managers and IT security practitioners, representing a balanced cross-section of organizations of varying sizes across multiple industries.

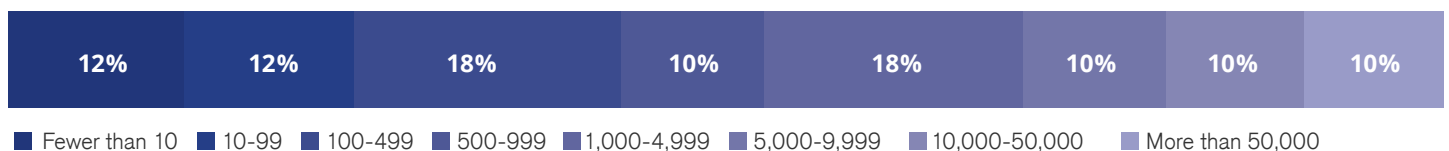
CAREER LEVEL



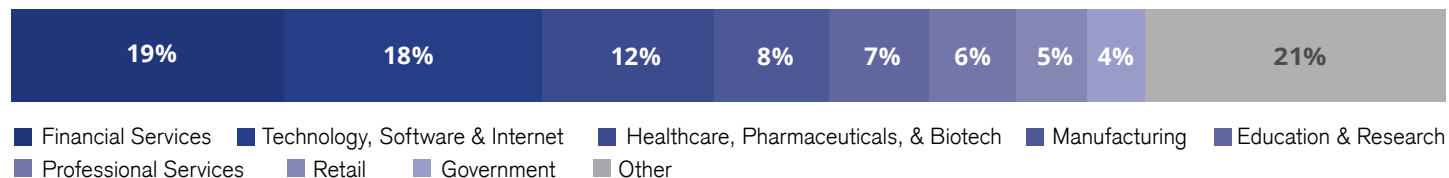
DEPARTMENT



COMPANY SIZE



INDUSTRY





SANGFOR

Making Each User's Digital Transformation Easier and More Secure

This is Sangfor Technologies' commitment to our customers. Since forming in 2000, Sangfor has been a global leader of IT infrastructure, security solutions, and cloud computing. Five business groups deliver industry-leading products for Hyper-Converged Infrastructure, Virtual Desktop Infrastructure, Next Generation Firewall, Secure Web Gateway, Endpoint Protection, Ransomware Protection, Incident Response, WAN Optimization, and SD-WAN. Constant innovation and dedication to creating value for our customers are the heart of our corporate strategy.

Sangfor's 7500+ employees take customers' business needs and user experience seriously by servicing and supporting them at over 60 branch offices globally in exciting locations like Hong Kong, Malaysia, Thailand, Indonesia, Singapore, Philippines, Vietnam, Myanmar, Pakistan, UAE, Italy, and the USA.

Sangfor has more than 100,000 satisfied customers worldwide, including Fortune Global 500 companies. Governments, universities & schools, financial institutions, manufacturing, and other industries trust us to protect them from the next generation of cyberthreats and help them on their journey to digital transformation with a future-proof IT infrastructure.

The Only Total Solution to Stop Ransomware

Sangfor's Security Solution for Ransomware is the only complete, holistic security solution to prevent and mitigate ransomware attacks in real-time. No other solution can impact every step in the ransomware kill chain and no other solution is modular enough to be tailored to the requirements and budget of the organization.

Sangfor Anti-Ransomware Solution

- Only solution that is proven to block every step in the ransomware kill chain
- Only solution with direct integration between firewall and endpoint agents and not using TI or management console as the go-between
- NGAF can block C2 communications and lateral propagation based on direct endpoint input
- NGAF can verify that the endpoint is infected based on C2 communications
- Only solution with a ransomware honeypot that effectively stops the encryption process and identifies the controlling application network-wide

For more information about Sangfor products & solutions, please visit our website at www.sangfor.com or send us an email to marketing@sangfor.com.